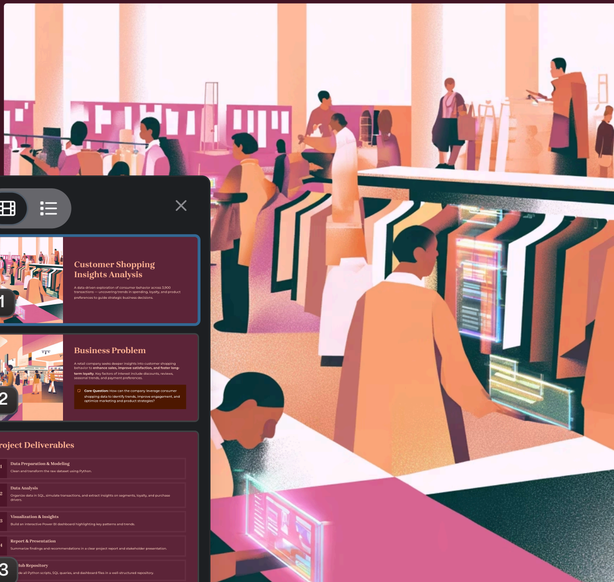
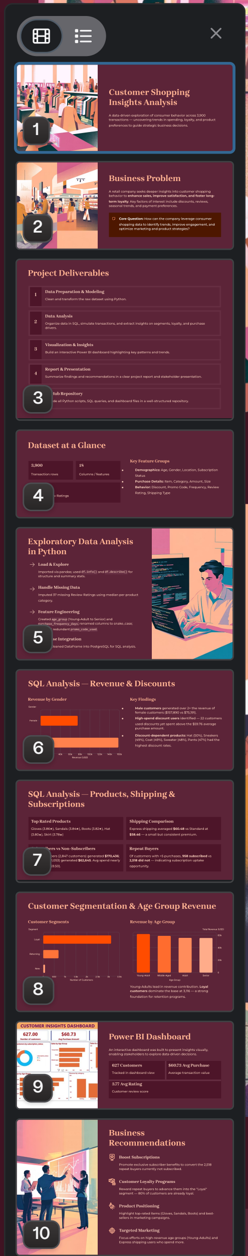


Customer Shopping Insights Analysis

A data-driven exploration of consumer behavior across 3,900 transactions — uncovering trends in spending, loyalty, and product preferences to guide strategic business decisions.



Business Problem

A retail company seeks deeper insights into customer shopping behavior to **enhance sales, improve satisfaction, and foster long-term loyalty**. Key factors of interest include discounts, reviews, seasonal trends, and payment preferences.

> **Core Question:** How can the company leverage consumer shopping data to identify trends, improve engagement, and optimize marketing and product strategies?

Project Deliverables

1

Data Preparation & Modeling

Clean and transform the raw dataset using Python.

2

Data Analysis

Organize data in SQL, simulate transactions, and extract insights on segments, loyalty, and purchase drivers.

3

Visualization & Insights

Build an interactive Power BI dashboard highlighting key patterns and trends.

4

Report & Presentation

Summarize findings and recommendations in a clear project report and stakeholder presentation.

5

GitHub Repository

Include all Python scripts, SQL queries, and dashboard files in a well-structured repository.

Dataset at a Glance

3,900

18

Key Feature Groups

- **Demographics:** Age, Gender, Location, Subscription

Exploratory Data Analysis in Python

→ Load & Explore

Imported via pandas; used `df.info()` and `df.describe()` for structure and summary stats.

→ Handle Missing Data

Imputed 37 missing Review Ratings using median per product category.

→ Feature Engineering

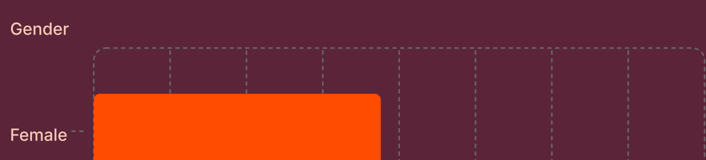
Created `age_group` (Young-Adult to Senior) and `purchase_frequency_days`; renamed columns to snake_case; dropped redundant `promo_code_used`.

→ Database Integration

Loaded cleaned DataFrame into PostgreSQL for SQL analysis.

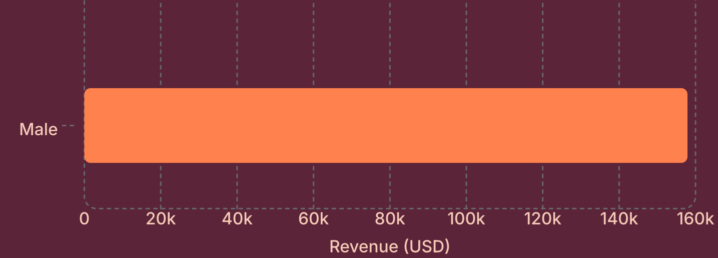
SQL Analysis — Revenue & Discounts

Revenue by Gender



Key Findings

- **Male customers** generated over 2× the revenue of female customers ($157,890 vs $75,191).



- **High-spend discount users** identified — 22 customers used discounts yet spent above the $59.76 average purchase amount.
- **Discount-dependent products:** Hat (50%), Sneakers (49%), Coat (49%), Sweater (48%), Pants (47%) had the highest discount rates.

SQL Analysis — Products, Shipping & Subscriptions

Top Rated Products

Gloves (3.86★), Sandals (3.84★), Boots (3.82★), Hat (3.80★), Skirt (3.78★)

Shipping Comparison

Express shipping averaged **$60.48** vs Standard at **$58.46** — a small but consistent premium.

Subscribers vs Non-Subscribers

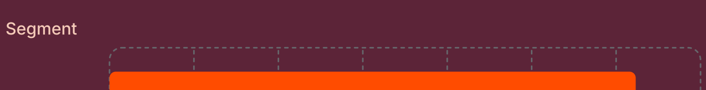
Non-subscribers (2,847 customers) generated **$170,436**; subscribers (1,053) generated **$62,645**. Avg spend nearly identical (~$59.50).

Repeat Buyers

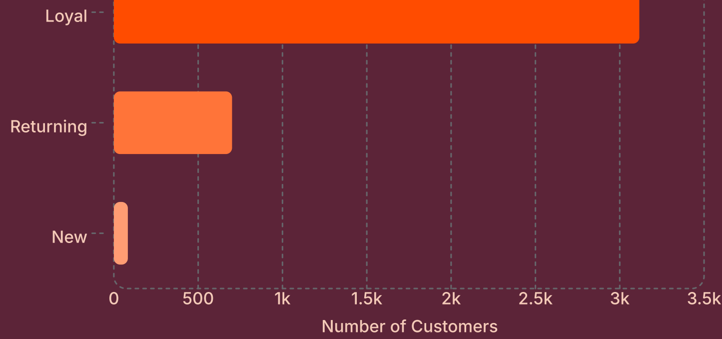
Of customers with >5 purchases, **958 subscribed** vs **2,518 did not** — indicating subscription uptake opportunity.

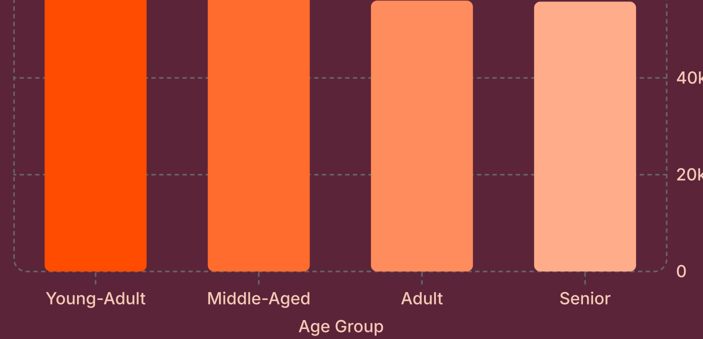
Customer Segmentation & Age Group Revenue

Customer Segments

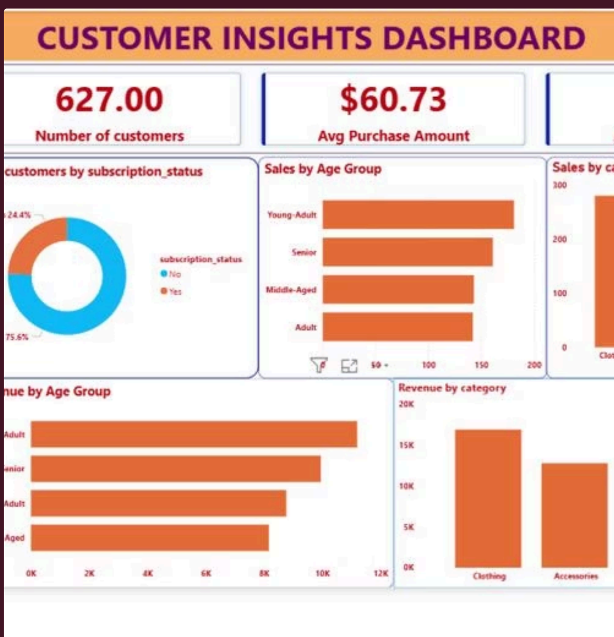


Revenue by Age Group





Young-Adults lead in revenue contribution. **Loyal customers** dominate the base at 3,116 — a strong foundation for retention programs.



Power BI Dashboard

An interactive dashboard was built to present insights visually, enabling stakeholders to explore data-driven decisions.

627 Customers Tracked in dashboard view	**$60.73 Avg Purchase** Average transaction value
3.77 Avg Rating Customer review score	

Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits to convert the 2,518 repeat buyers currently not subscribed.

Customer Loyalty Programs

Reward repeat buyers to advance them into the "Loyal" segment — 80% of customers are already loyal.

Product Positioning

Highlight top-rated items (Gloves, Sandals, Boots) and best-sellers in marketing campaigns.

Targeted Marketing

Focus efforts on high-revenue age groups (Young-Adults) and Express shipping users who spend more.